

INMO(22nd) – 2007

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. In a triangle ABC right angled at C, the median through B bisects the angle between BA and the bisector of $\angle B$. Prove that.
 2. Let n be a natural number such that $n = a^2 + b^2 + c^2$, for some natural numbers a, b, c . Prove that: $9n = (p_1a + q_1b + r_1c)^2 + (p_2a + q_2b + r_2c)^2 + (p_3a + q_3b + r_3c)^2$ where p_j, q_j, r_j are all nonzero integers. Further if 3 does not divide at least one of a, b, c , prove that $9n$ can be expressed in the form of $x^2 + y^2 + z^2$, where x, y, z are natural numbers none of which is divisible by 3.
 3. Let m and n be the positive integers such that the equation $x^2 - mx + n = 0$ has real roots α, β . Prove that α, β are integers if and only if $[m\alpha] + [m\beta]$ be the square of an integer. (Here $[x]$ is the greatest integer function.)
 4. Let $\sigma = (a_1, a_2, \dots, a_n)$ be a permutation of $(1, 2, 3, \dots, n)$. A pair (a_i, a_j) is said to correspond to an inversion of σ , if $i < j$ but $a_i > a_j$. (Example: in the permutation $(2, 4, 5, 3, 1)$, there are 6 inversions corresponding to the pairs $(2, 1), (4, 3), (4, 1), (5, 3), (5, 1), (3, 1)$.) How many permutations of $(1, 2, 3, \dots, n)$, $n \geq 3$, have exactly two inversions.
 5. Let ABC be a triangle in which $AB = AC$. Let D be the midpoint of BC and P be a point on AD. Suppose E is the foot of the perpendicular from P on AC. If $\frac{AP}{PD} = \frac{BP}{PE} = \lambda$, $\frac{BD}{AD} = m$ and $z = m^2(1 + \lambda)$, prove that $z^2 - (\lambda^3 - \lambda^2 - 2)z + 1 = 0$. Hence show that $\lambda \geq 2$ and $\lambda = 2$ if only if ABC is equilateral.
 6. If x, y, z are positive real numbers, prove that :

$$(x + y + z)^2 (yz + zx + xy)^2 \leq 3(y^2 + yz + z^2)(z^2 + zx + x^2)(x^2 + xy + y^2).$$

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